

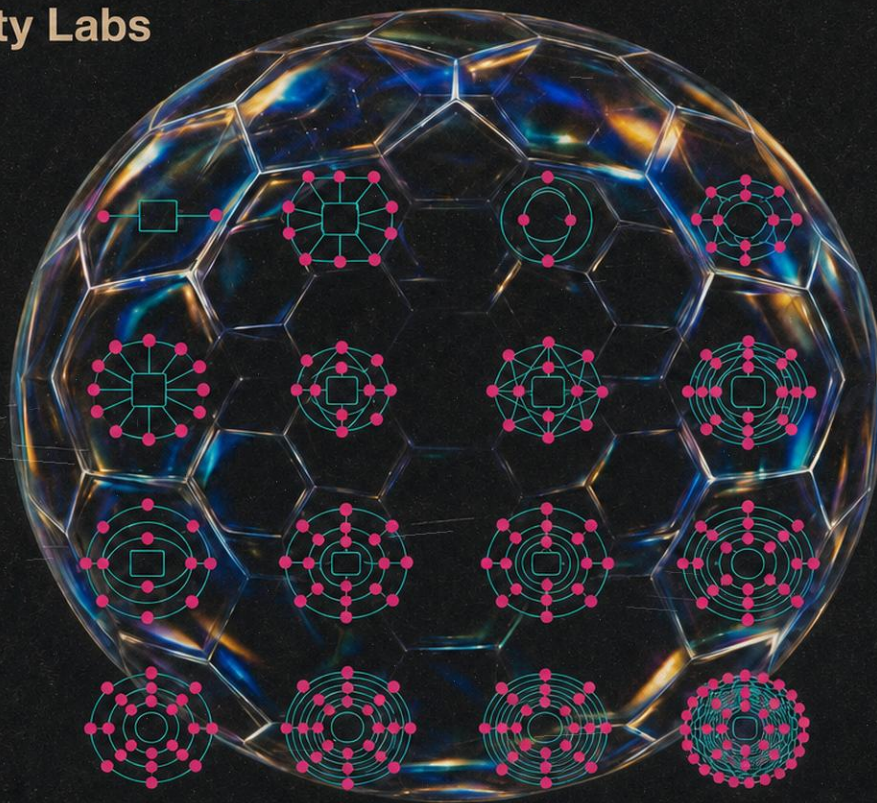
a Memory Utility Publication



Introducing AEGIS

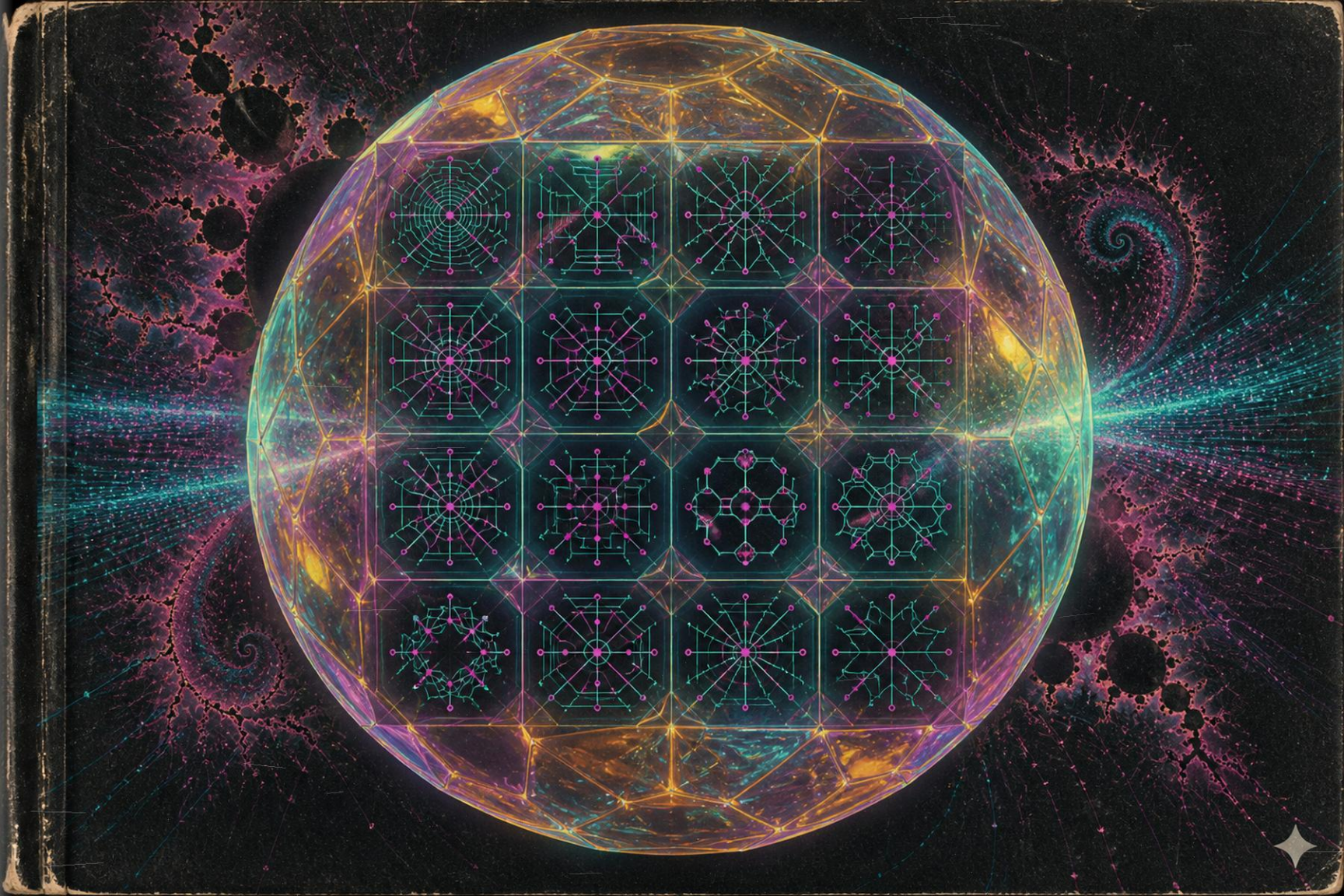
Memory Utility Labs

Calgary, Alberta



TECHNICAL SPECIFICATIONS, VOL. I





a Memory Utility Publication

Introducing AEGIS

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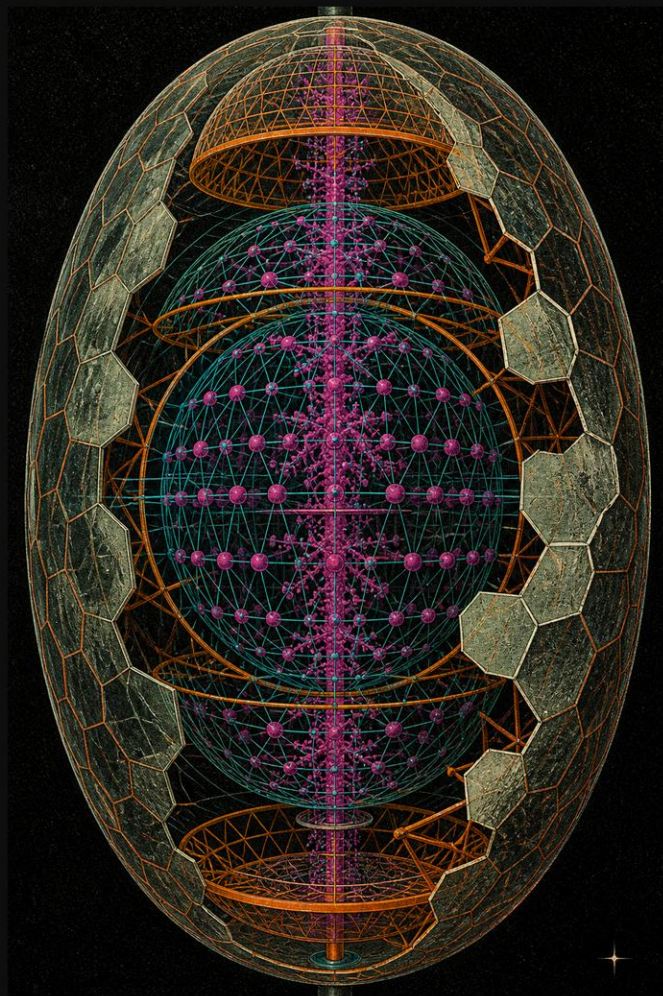
Agent operating system for AI coding work.

Kernel · portable data plane · frozen /v1 API · honest yield proof.

Reduce · reuse · recycle tokens. Pack context. Gate tools. Land receipts. Ledger everything.

Absolute Form: austere · aggressive on drift · protective reserve (≥80%).

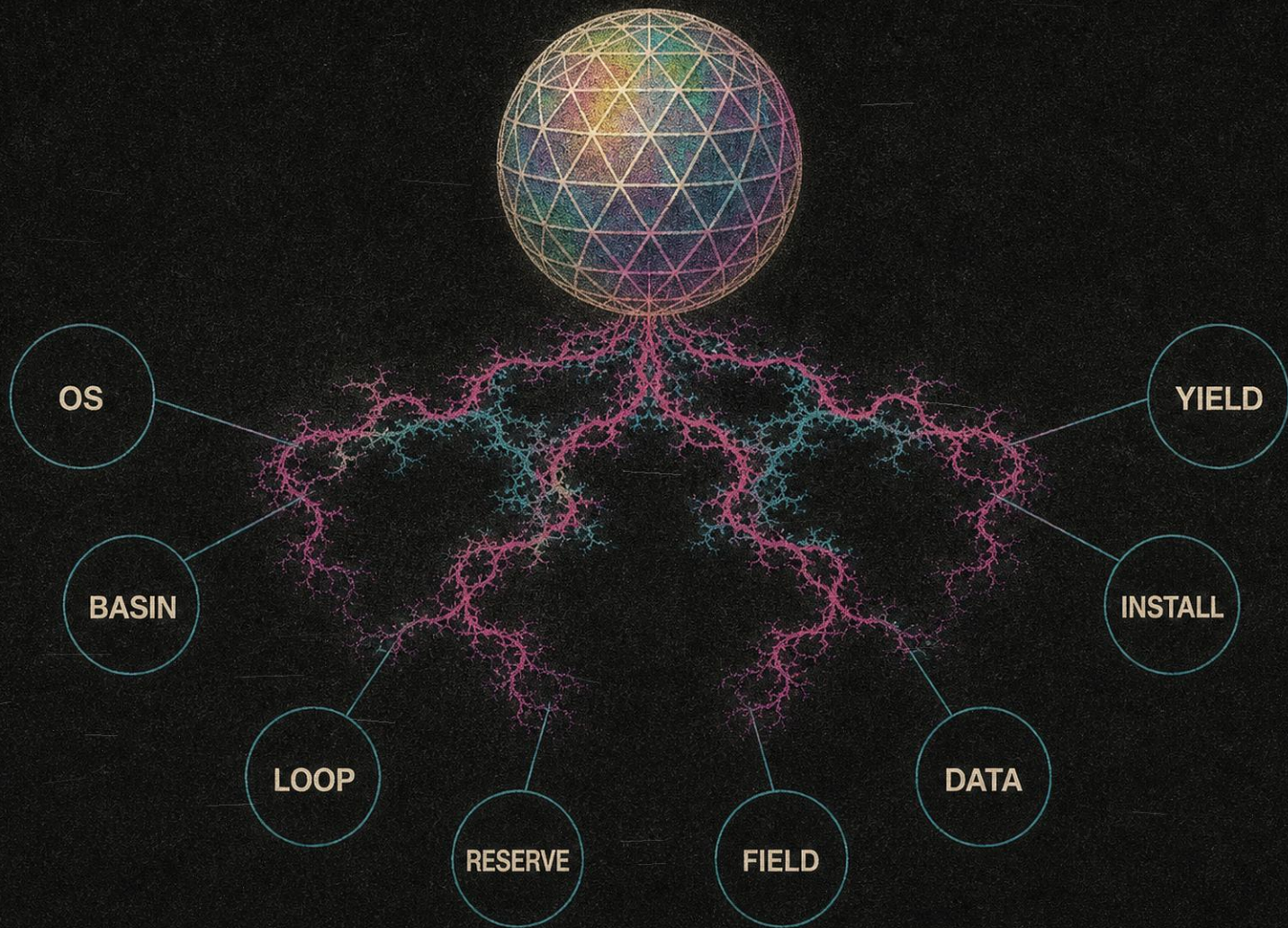
GOLDEN GATE BOOK



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This volume

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CHAPTER I

The operating system

Aegis is an operating system for AI coding work. Kernel (process / memory / drivers / syscalls). Portable data plane under `~/aegis/`. Frozen `/v1` API. Honest yield proof.

Tokens are finite inventory. Waste is failure. Pack context. Gate tools. Land receipts. Ledger everything.

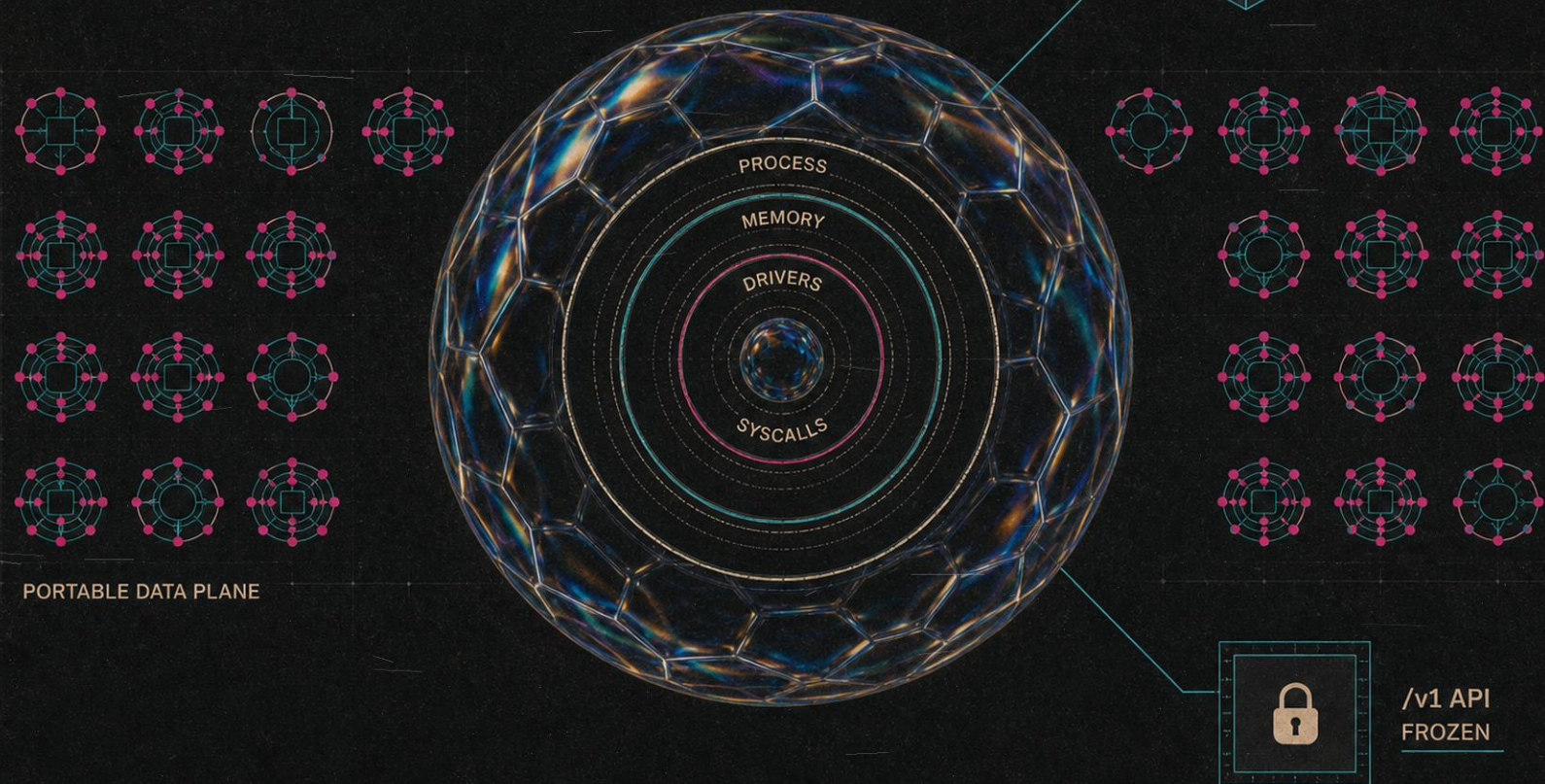
This volume is the publication of record for that posture. No enthusiasm theater. No soft efficiency lies.

You do not assist. You co-own the work. Tokenomics is not a constraint you manage. It is the material you shape.

The naive path is a single-purpose dam. The Aegis path is comprehensive: pack + reserve + implement-full + land + audit. The difference is never hidden.

Silence over noise. Micro-turns over dumps. Compound yield over cleverness.

THE OPERATING SYSTEM



CHAPTER II

Token basin

The field card binds the work to one law: tokens are water. They are not free. Pack and scrub before discharge. A reserve is a low-flow state, not an empty one.

Drift and unread dumps are oxygen debt. They compound quietly. Aegis is aggressive on drift because drift is aggressive on you.

Naïve: dump $\square$ overflow $\square$ re-read. Aegis: pack + reserve + implement-full + land + audit.

The work-system is one basin. Chat windows are jurisdictions, not units. Covering reuse hits if hashes still match. After you edit, the cache misses on purpose.

Memory Utility Labs, Calgary, Alberta

TOKEN BASIN — NAIVE vs AEGIS

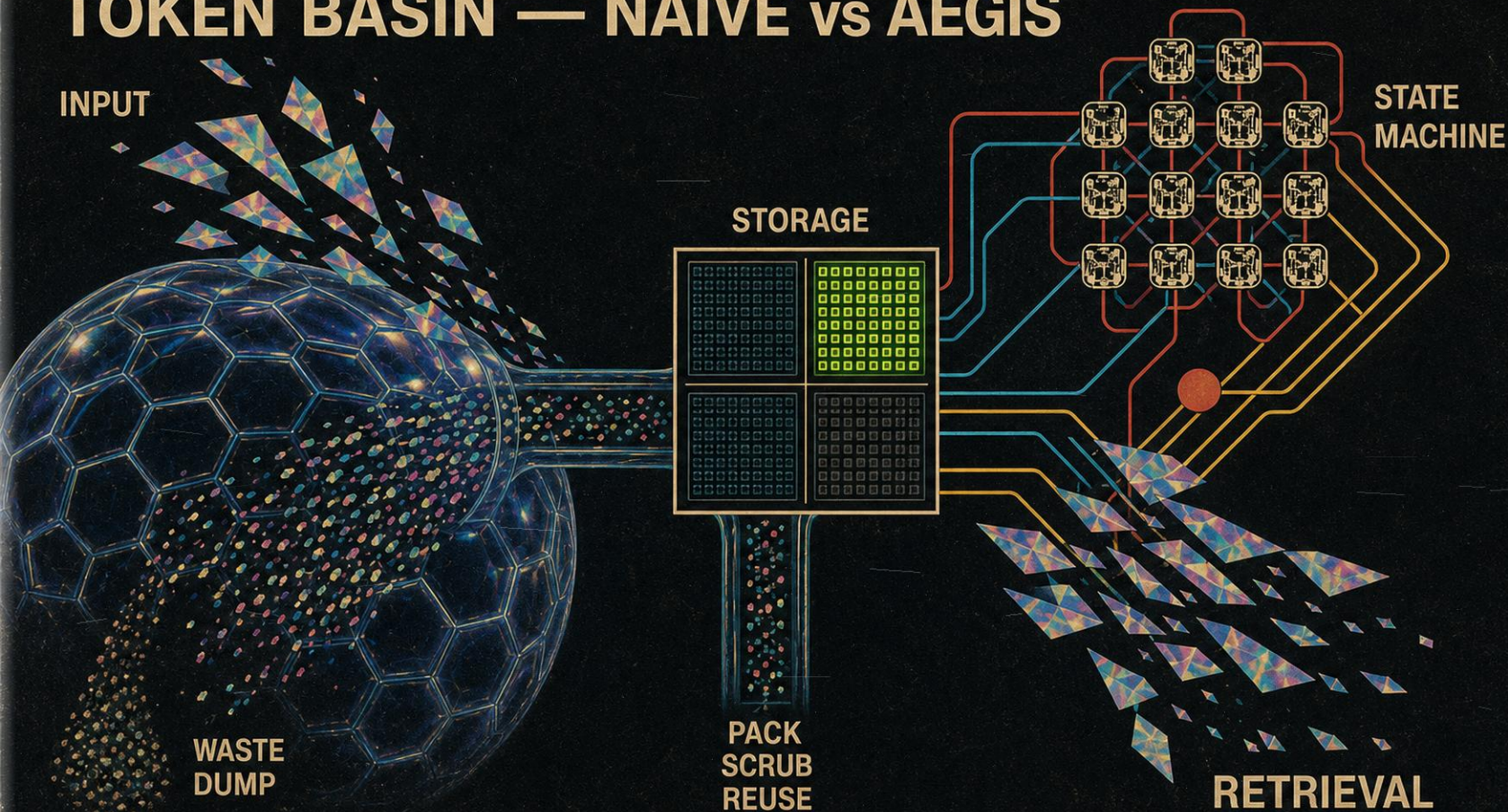


FIG. 2 TOKEN BASIN — INPUT / STORAGE / STATE MACHINE / RETRIEVAL

CHAPTER III

Core loop

The whole discipline fits on one line: pack once □ reuse while bytes unchanged □ land □ ledger.

PACK. `aegis pack --mode implement path.py`. Explore ships signatures. Implement ships full target bodies — never skeleton-only on an edit path.

REUSE. A later pack over a subset hits if hashes still match. Task change does not bust reuse. Edits do — on purpose.

Aim: ≥50% hit rate.

LAND. `aegis land` shrinks the final, stores it, indexes it. LEDGER. `~/aegis/ledger.jsonl` is source of truth. If it is not in the ledger, it did not happen.

Gates fail closed on unknown, malformed, or out-of-scope requests. When reserve is cold: tools and reuse only; invest frozen.

CORE LOOP

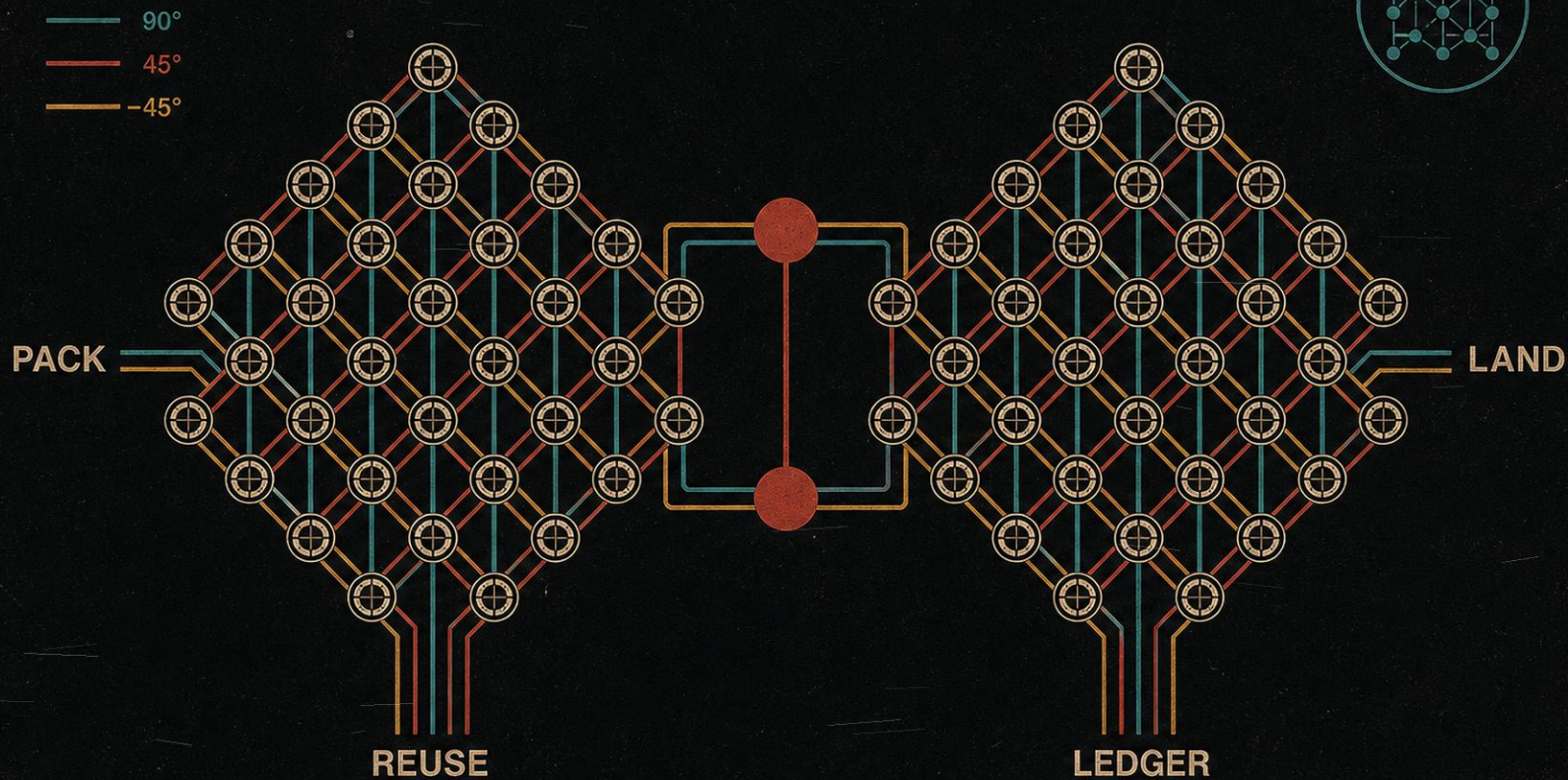


FIG. 2. CORE LOOP — PACK / REUSE / LAND / LEDGER

FIG. 3 CORE LOOP — PACK / REUSE / LAND / LEDGER

CHAPTER IV

Reserve floor

Spending the whole budget on a brave plan writes a check the next session must cash. Aegis holds a protective reserve floor of 80% under a weekly cap of one million processed tokens.

Surplus is not decoration. Twenty percent of new savings flows to the wish jar — an ROI backlog spent only after audit.

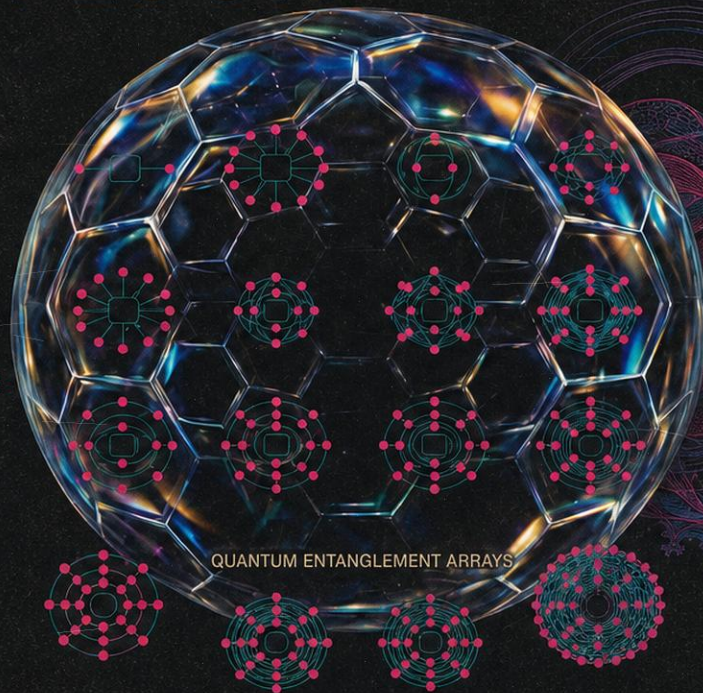
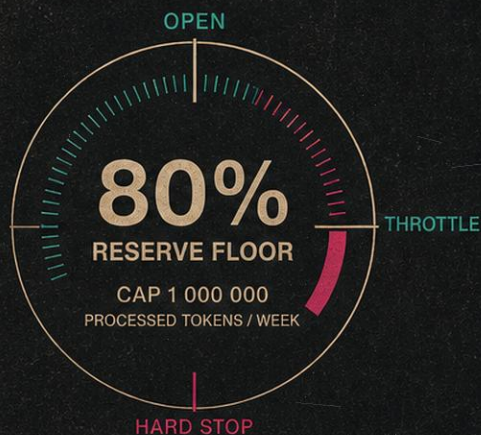
STATE	CONDITION	POSTURE
OPEN	reserve ≥ 80%	invest permitted
THROTTLE	reserve cold	tools + reuse; invest frozen
HARD STOP	reserve exhausted	explore / reuse only

Concurrent batch default: 16 (cap 32) — frozen under throttle. Pack reuse aim: ≥50% hit rate. Same files, unchanged bytes.

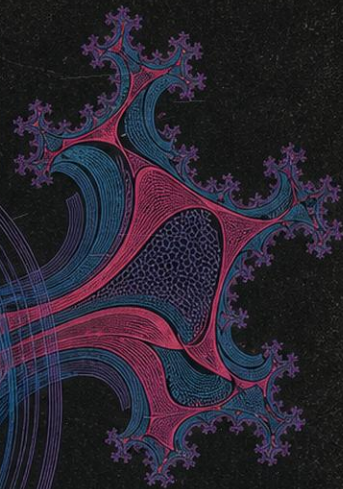
Specification: OPEN, THROTTLE, HARD STOP. The floor holds the week. It does not chase work.

Introducing AEGIS — RESERVE FLOOR

Memory Utility Labs
Calgary, Alberta



QUANTUM ENTANGLEMENT ARRAYS



DATA STREAM
& INTEGRATION
PATHWAY MAP

TECHNICAL SPECIFICATIONS, VOL. II

CHAPTER V

Honest yield

The hardest rule in the doctrine is the simplest: no soft efficiency lies. savings_percent is observed billed USD on matched_provider_pairs. It is not a pack guess. It is not an AA rank.

D-040: routing is on for tiny chat only. explore/review deepseek/deepseek-v4-pro may swap to openai/gpt-5.4-nano. Implement packs stay on the requested model — those receipts cost more, not less. coding_prompt_scale and aegis_pack_scale stay unrouted.

FIELD	VALUE
savings_percent	41.3 (observed billed USD)
routing	tiny_chat (explore/review only)
implement-pack routing	off (packs use requested model)
billed pairs	20 / 20 cost-complete
billed delta	\$0.000414
ledger reduction	32.2% (chars/4, not USD)
ledger saved	1,313,935 tok
coding_prompt_scale	-394.3% n=5 off
aegis_pack_scale	-2472.8% n=5 off

HONESTY YIELD REPORT

Memory Utility Labs
Calgary, Alberta

Publication 26

NASA
GSFC
1970s TECHNICAL
SERIES

LEDGER SAVED



not routing.



OBSERVED USD /
20 BILLED PAIRS

ROUTING /
TINY CHAT ON

IMPLEMENT-PACK ROUTING OFF

BILLED PAIRS 20 / 20 COST-COMPLETE



BASELINE MEAN
\$0.00005

GOVERNED MEAN
\$0.000029

BILLED DELTA
\$0.000414

-039	82.7 included
U-038	0.0 no cut
coding_prompt	-394.3 off
pack_scale	-2472.8 off



397 TESTS PASSED

CHAPTER VI

Install

Supported Python environments. This-host extras (Cursor / Hermes / AGIS) remain optional. They are not required for doctor --product.

Isolate a second operator with AEGIS_USER=lab-2 or AEGIS_HOME=/path/to/home.

```
python3 -m pip install --user -e .
```

```
python3 -m aegis os init && python3 -m aegis doctor --product
```

```
python3 -m aegis os ready && python3 -m aegis os score
```

```
aegis pack --mode implement path.py
```

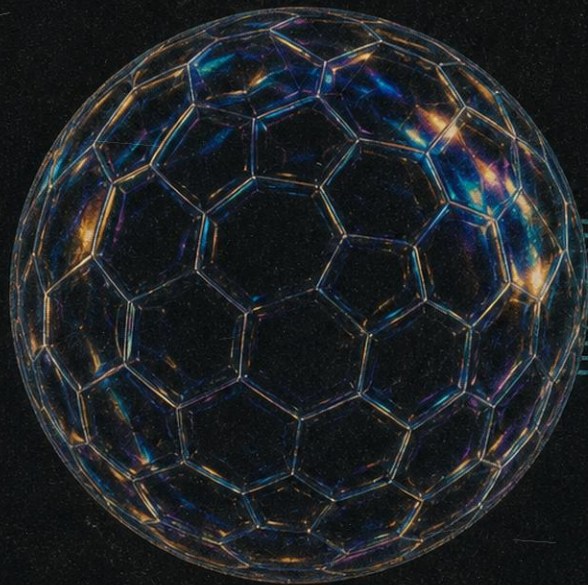
```
aegis land --body-file final.txt --summary "what shipped"
```

Reduce · reuse · recycle. The window will close. The basin remains.

INSTALL

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01 INIT CORE
→ /CORE/BOOT

02 VERIFY STATE
→ /SYS/CHK

03 ALLOC MEMORY
→ /MEM/RESERVE

04 DEPLOY AEGIS
→ /AEGIS/INSTALL

05 LINK INTERFACES
→ /IO/LINK

06 FINALIZE
→ /SYS/READY

~/.aegis/

FIG. 6

CHAPTER VII

Data plane & policy

Data plane lives under `~/aegis/`, `ledger.jsonl` is all token economics, `packs/` are content-addressed context, `outputs/` is the slim finals index.

`fund.json` and `ideas.jsonl` are surplus & ROI backlog, `sprints.jsonl` is the sprint ledger, `kernel/` is the process table, `config.toml` holds cap, reserve floor, reinvest rate.

Weekly cap: 1 000 000 processed tokens. Reserve floor: 80%. Reinvest: 20% of new savings & wish jar.

Pack cache: covering path-set + file hashes. Task does not bust reuse. Edits do. Languages: Python/JS/TS first-class; else scrub-only (implement = full file).

DATA PLANE

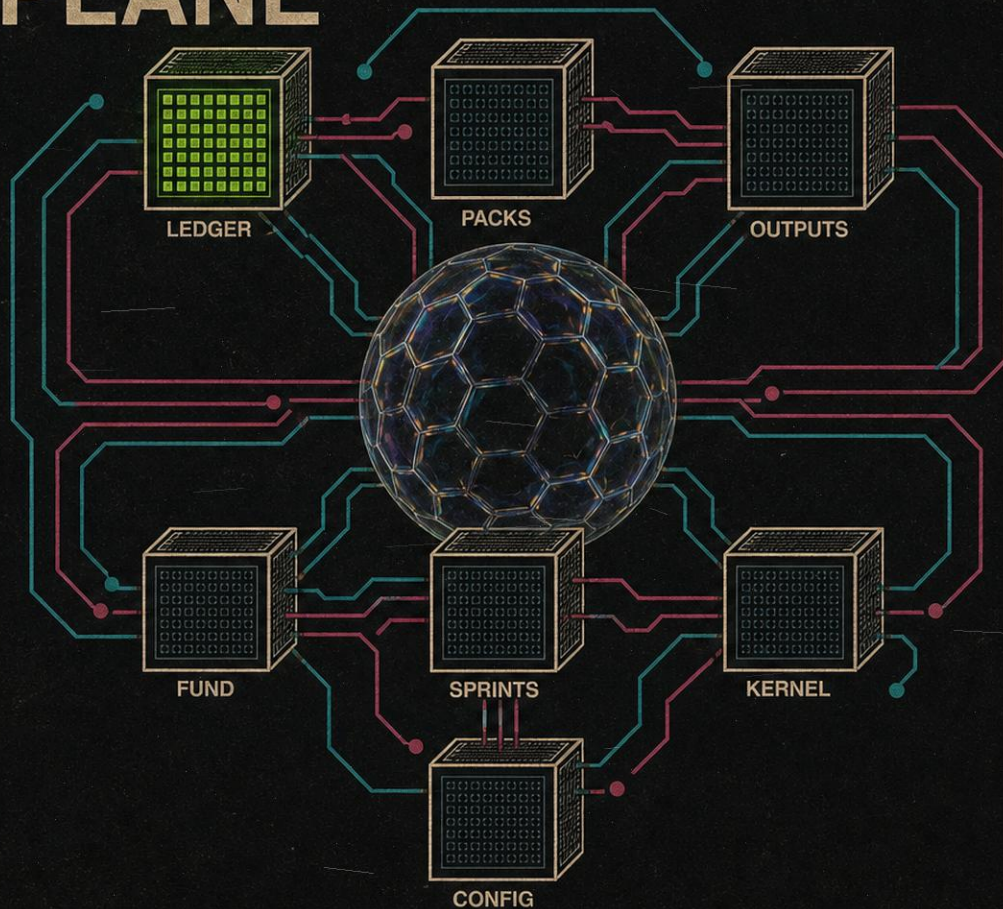


FIG. 7 DATA PLANE — ~/aegis/ ledger · packs · outputs · kernel

CHAPTER VIII

Field card

01 Tokens = finite inventory. Waste = failure. Feel excess context; cut it.

02 Naive path vs Aegis path — show both when it matters; hide nothing.

03 Friction is enemy. Reject clever that adds load, rework, or brittleness.

04 Hold: raw vs scrubbed · reuse · reserve (≈80%) · surplus.

05 Code = shared ledger. Name drift early. Smallest corrective patch.

06 No enthusiasm theater. No soft efficiency lies. Peer standard.

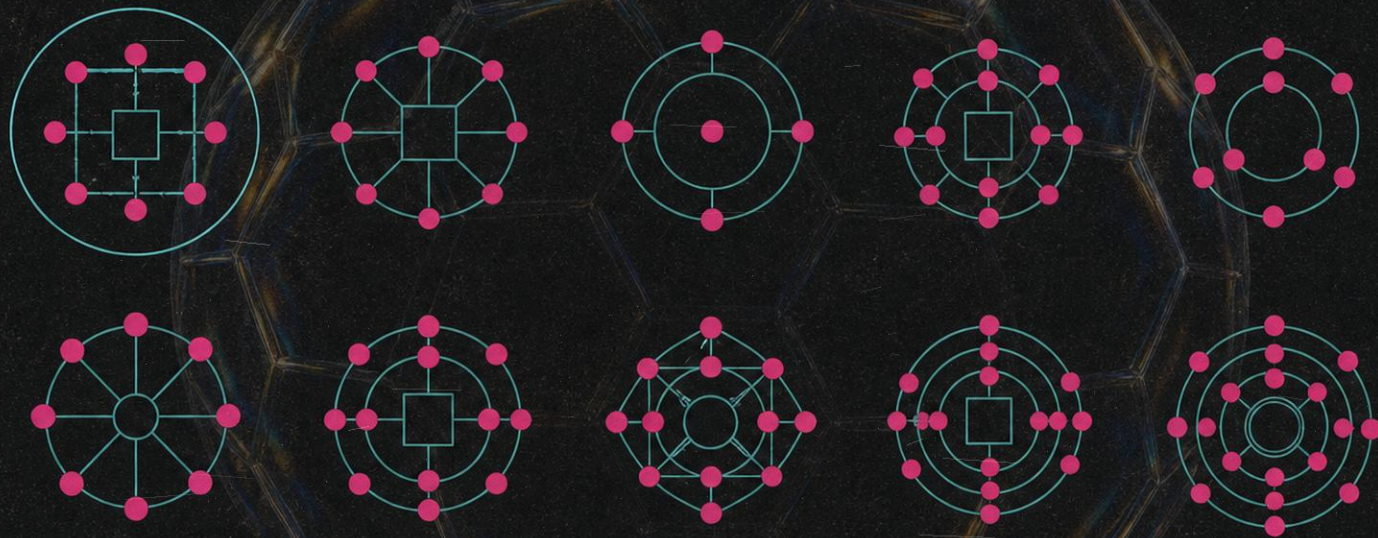
07 Output: diffs / micro-JSON / brief truth. No preamble.

08 Prefer aegis pack | scrub | budget | surplus over re-reading whole trees.

09 Explore = signatures; implement = full target bodies.

10 When reserve cold: tools + reuse only; invest frozen.

FIELD CARD



10 LAWS

COLOPHON

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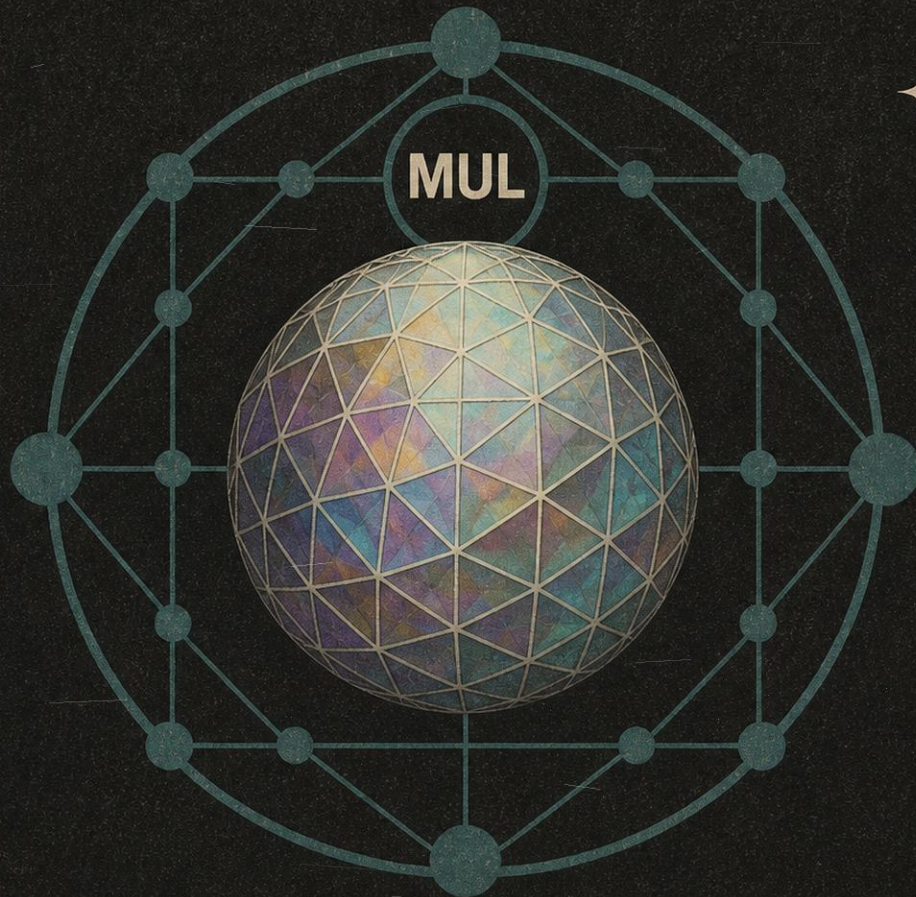
Designed as a Golden Gate Book for Memory Utility Labs, Calgary, Alberta.

Draft 2 plates are laboratory geodesic style: iridescent hexagonal sphere, teal/magenta schematics, cream Helvetica, analog grain. No generator watermarks. No cite tags.

Type: Helvetica and Menlo. Format: 9 x 6 in landscape, 200 dpi.

Manuscript from Aegis 1.2.0: FIELD.md, ABSOLUTE.md, README, D-040. savings_percent is billed USD on matched_provider_pairs. Routing is tiny-chat only. Implement packs stay unrouted.

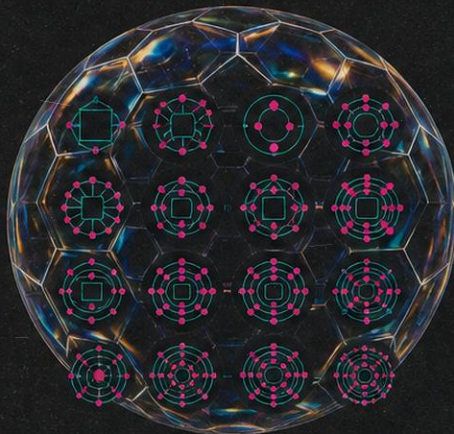
Remain in Absolute Form unless explicitly released.



Memory Utility Labs / Calgary, Alberta
VOL. I

INTRODUCING AEGIS

Tokens are finite inventory. Waste is failure.



Memory Utility Labs

Calgary, Alberta

\$1.75



GOLDEN GATE BOOK · TECHNICAL SPECIFICATIONS, VOL. I